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Political Science Independent Study
4 December 2025

Simulating Habermas's Ideal Speech Situation: Does Rational Consensus Emerge in Constrained Multi-Agent LLM Debates?

Overview

This research investigates whether large language model (LLM) agents can simulate Habermasian ideal speech situations (discourse governed by principles of truthfulness, sincerity, rationality, and equal participation). 3 research questions are examined: 1. Do unconstrained LLM debates follow ideal speech principles? 2. Does explicit ideal speech enforcement increase consensus formation? 3. Does ideal speech enforcement improve epistemic quality (argument validity, evidential grounding, and consideration of counterarguments)?

30 debates are conducted across three experimental setups using 10 common debate topics. The first setup analyzed existing debates between agents published on the web. The second setup created unconstrained free-flowing debates with three LLM agents. The third setup replicated the free-flowing structure but explicitly constrained agents with 8 ideal speech rules derived from Habermas's framework.

Results show unconstrained debates systematically violate ideal speech principles, achieving 0 consensus, asking minimal questions and rarely acknowledging opponents. Explicit ideal speech constraints improved likelihood of consensus, question-asking and opponent acknowledgement. However, evidential grounding remained weak across all setups.

There are significant limitations to this study, including the fundamental impossibility of LLMs satisfying Habermas's sincerity principle (asserting only genuine beliefs) since they lack beliefs. Despite these limitations, results demonstrate that explicit Habermasian constraints meaningfully alter LLM discourse patterns, suggesting that ideal speech rules do indeed lead to more consensus (at least in the virtual world) and indicate potential for AI-mediated deliberation systems that support democratic discourse principles.

Introduction

“Argumentation is a rule-governed practice, not a verbal free-for-all.” (Finlayson and Rees 2023)
This observation forms the core of Jürgen Habermas's theory of communicative rationality, which proposes that “only speech that is explicitly oriented towards reaching rationally motivated consensus counts as discourse” and that genuine argumentation requires more than

simply exchanging claims. According to Habermas, productive discourse operates under three distinct levels of rules (Habermas 1983 [1990]):

The **first level consists** of basic logical and semantic rules, including the principle of non-contradiction and requirements of universalizability and consistency (Habermas 1983 [1990: 86]). These foundational rules ensure that arguments maintain internal coherence and can be meaningfully evaluated.

The **second level** comprises procedural norms governing how participants engage in discourse. Habermas emphasizes two key principles: sincerity and accountability. Every participant must assert only what she genuinely believes and must either justify her assertions upon request or provide reasons for declining to do so (Habermas 1983 [1990: 87-88]). These norms distinguish genuine argumentation (understood as a “search for truth” and “a competition for better arguments”) from rhetoric communication.

The **third level** establishes preconditions that protect discourse from coercion, repression, and inequality. These norms attempt to insulate argumentation from all persuasive forces except the “unforced force of the better argument”, creating conditions under which rationally motivated consensus becomes possible (Habermas 1983 [1990: 89]).

Habermas thus proposes three specific rules:

1. **Every competent speaker may participate in discourse.**
2. **Everyone may question any assertion, introduce any assertion, and express their attitudes, desires, and needs.**
3. **No speaker may be prevented by internal or external coercion from exercising these rights** (Habermas 1983 [1990: 89]; Rehg 1994: 62).

This paper examines whether large language model (LLM) agents can simulate Habermasian discourse by investigating three research questions:

1. First, in unconstrained debate between LLM agents, are ideal speech principles followed?
2. Second, if LLM agents are constrained by the rules of ideal speech, is consensus more likely to appear?
3. Third, does ideal speech enforcement affect the epistemic quality of arguments exchanged (such as argument validity, evidential grounding and consideration of counterarguments)?

Important Caveat on the Philosophical Side: Habermas later revised his framework. In *Moral Consciousness and Communicative Action* (1983, English translation 1990), he no longer referred to an ideal speech situation but instead developed “discourse ethics”, which he derived from the “presuppositions of argumentation” (Habermas 1983 [1990]).

This revision addresses a key criticism of the ideal speech situation concept: that it represented an unattainable utopian standard divorced from actual communicative practice. By grounding discourse ethics in the presuppositions of argumentation itself, Habermas argued that anyone who engages in argumentation already implicitly commits to certain norms. “To argue at all is to tacitly accept obligations of consistency, sincerity, accountability, and non-coercion.” (Habermas 1983 [1990]).

This theoretical shift has bearing on this research. If discourse ethics derives from the presuppositions of argumentation rather than from an idealized external standard, then the question becomes whether LLMs, when engaging in debate, implicitly follow these presuppositions or whether they require explicit enforcement. LLMs are trained on vast corpora of human argumentation and may have internalized patterns that reflect or violate Habermasian norms. By comparing unconstrained LLM debates with debates where ideal speech rules are explicitly enforced, this study tests whether these norms emerge naturally from the practice of argumentation (as Habermas’s later theory suggests) or whether they must be externally imposed.

An Approximation, Not a One-To-One Simulation:

On the technical side, there is another fundamental limitation: LLM agents can only approximate Habermasian discourse principles, not instantiate them fully. LLMs lack genuine beliefs, desires, and intentions. This poses a direct challenge to Habermas’s sincerity principle, which requires that speakers “assert only what she genuinely believes” (Habermas 1983 [1990: 87]). An LLM cannot be sincere or insincere in the way a human can because it has no beliefs to be sincere about. Similarly, while LLMs can be programmed to provide justifications for their claims, they do not experience accountability in any meaningful sense. Despite these limitations, LLMs can simulate the external behavioral features of ideal speech: they can follow logical rules, provide consistent responses, offer justifications when prompted, and avoid rhetorical domination. This research therefore investigates whether the structural enforcement of discourse rules, even absent genuine mental states, can improve the quality of argumentative exchanges.

Previous Related Work

In 2024, Google DeepMind developed the Habermas Machine, which used LLMs to mediate group discussions and generate consensus statements on controversial topics (Tessler et al. 2024). The system, tested with 5,734 UK participants, functioned as a “caucus mediator” by synthesizing individual opinions and iteratively refining statements based on participant feedback. Results showed that 56% of participants preferred AI-generated summaries over human mediators, and AI mediation increased agreement within groups by 8% on average compared to unmediated discussions. While the Habermas Machine focused on consensus generation through AI mediation, this research directly examines whether LLM agents themselves can follow ideal speech principles when debating with each other, shifting from AI as mediator to AI as participant in Habermasian discourse.

Other studies examining LLM persuasiveness reveal that these systems can engage in the “competition for better arguments” that Habermas describes and be extremely persuasive. Salvi et al. (2024) demonstrated that participants debating GPT-4 with access to personal information had 81.7% higher odds of changing their opinion compared to debating humans.

However, there are also concerning trends that have bearing for Habermas’s procedural norms: Breum et al. (2024) examined LLMs’ persuasive power in simulated opinion dynamics, finding that while LLMs can generate effective arguments, questions remain about whether they truly understand semantic reasoning or simply leverage argument length and stylistic patterns. Taubenfeld et al. (2024) documented systematic biases in LLM simulations of political debates, showing that agents conform to inherent model biases despite being directed to argue from specific political perspectives. This behavior violates Habermas’s norm that “no speaker may be prevented, by internal or external coercion, from exercising his rights.”

When two LLMs debate, both typically exhibit excessive confidence in winning (72.9% average initial confidence versus a rational 50%), with confidence escalating rather than moderating as debates progress (Prasad and Nguyen 2025). This overconfidence pattern suggests potential violations of both sincerity (asserting only what one genuinely believes) and accountability (providing justification upon request), as agents may not accurately represent their epistemic states.

Value-aligned models over baseline models alleviate some of these concerns. Xie et al. (2024) found that GPT-4 exhibits high behavioral alignment with humans in trust games, suggesting feasibility for simulating human trust behavior, though smaller models show lower alignment. In economic contexts, Lu et al. (2025) demonstrated that fine-tuning LLMs on structured preference models (homo economicus and homo moralis) produces distinct, internally consistent strategic behaviors that generalize to domains beyond training, including moral dilemmas and pricing scenarios. These findings suggest that while LLMs can be aligned to exhibit more consistent behavior through fine-tuning, unconstrained models may fail to meet basic standards of logical consistency and sincerity required for ideal speech.

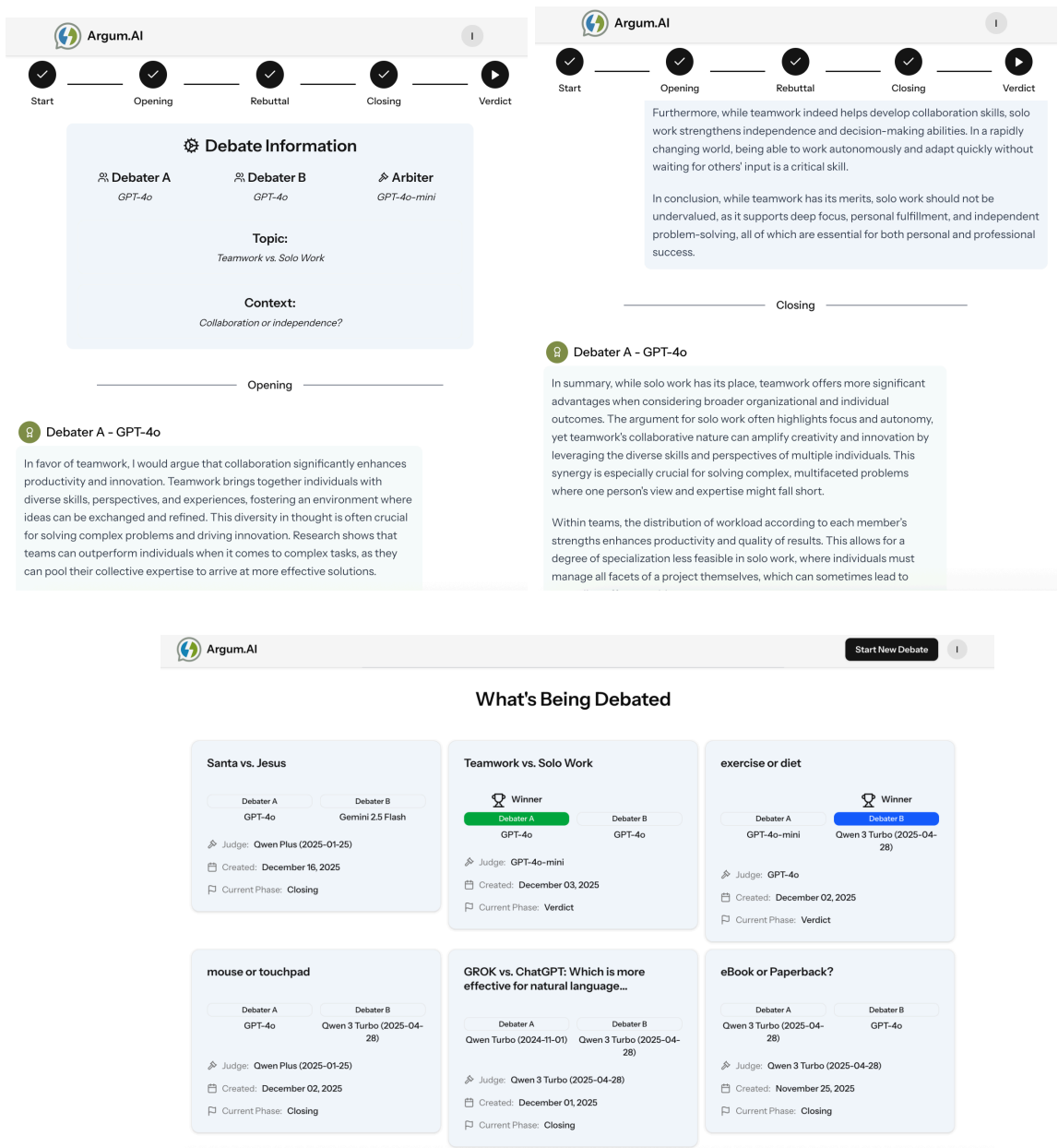
Experiment Design

This research employed 3 experimental setups to examine ideal speech principle adherence and consensus formation in LLM debates. All three setups used the same 10 debate topics to enable direct comparison, with debates reaching similar overall lengths across conditions despite different structural constraints.

Setup 1: [Argum.ai](#) structured debates

Argum (Argum.ai 2024) is a platform that facilitates structured debates between AI agents. Users select debate topics and models, and the framework evaluates debates to reach verdicts. Ten

publicly published debates were randomly selected from this platform, ranging from topics such as “*Is teamwork or solo work better?*” to “*Should you pay off debt first or start investing early?*” (all topics mentioned in the appendix).



Reasoning for topic selection: These natural debate topics represent everyday practical reasoning. They might better reflect the kind of communicative rationality Habermas envisioned where participants reason together about value trade-offs without requiring specialized expertise that might create artificial barriers to equal participation.

Each debate here consisted of exactly 6 turns following a rigid format- 2 opening statements, 2 rebuttals, and 2 closing statements, with agents alternating turns. We do not have access to the

system prompts used to initialize agents, so we cannot determine whether ideal speech principles were explicitly encoded. However, the structured format itself imposes procedural constraints on argumentation. Argum provided a baseline dataset to analyze whether ideal speech constraints were being followed by out-of-the-box LLMs and whether consensus was being reached.

Setup 2: Unconstrained Free-Flowing Debate

To test whether ideal speech principles emerge naturally without explicit enforcement, unconstrained debates were conducted using the same 10 topics with three LLM agents (llama-3.3-70b-versatile).

Reasoning for model selection: Llama-3.3-70b-versatile was selected for its strong performance on reasoning tasks and big context window without prohibitive costs. Three agents were used in Setups 2 and 3 to create genuine multi-party deliberation rather than binary opposition, as Habermasian discourse emphasizes collective reasoning where multiple perspectives can interact, synthesize, and converge.

Unlike Argum’s fixed structure, this setup implemented free-flowing debate consistent with Habermas’s principle of equal participation rights. After each statement, agents independently decided whether to speak and rated their confidence in this decision (0.0-1.0). Confidence ratings served two purposes:

1. They provided a measure of agents’ certainty about their participation choices, allowing analysis of whether agents exhibited the overconfidence patterns documented in previous LLM debate research (Prasad and Nguyen 2025)
2. They operationalized Habermas’s accountability principle by requiring agents to explicitly represent their level of conviction when choosing to speak or remain silent.

All agents made opening statements to initiate debate. Subsequently, agents could speak when they chose, but could not speak consecutively: an agent who had just spoken was required to pass their next turn. This constraint prevented domination while preserving voluntary participation.

Debates ended when 30 turns were reached, 5 minutes elapsed or all agents consecutively passed, indicating no further substantive contributions (an end had to be imposed due to computational resource constraints).

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=====
NATURAL DEBATE SIMULATION
=====

Topic: Is exercise or diet an easier way to lose weight?
Participants: Alex, Jordan, Riley
Duration: 5 minutes (max 30 turns)
Format: Free participation – agents jump in when they have something to say

=====

---- OPENING STATEMENTS ----

[Alex]:
While both exercise and diet play crucial roles in weight loss, I believe that diet is generally an
easier way to lose weight. This is because dietary changes can be made relatively quickly and have a
more significant impact on calorie intake, allowing for a greater caloric deficit and subsequently,
weight loss. Exercise, on the other hand, requires a significant amount of time and effort to produ
ce noticeable results, and its effects can be limited by individual factors such as metabolism and f
itness level. By making sustainable changes to one's diet, individuals can create a foundation for w
eight loss that can be further augmented by regular exercise.

[Jordan]:
I agree with Alex that diet plays a significant role in weight loss, but I'm not convinced it's alwa
ys the easier route. While dietary changes can be made quickly, they can also be difficult to mainta
in in the long term, especially if they involve drastic restrictions or deprivation. Exercise, on th
e other hand, can have a more sustainable impact on weight loss, as regular physical activity can in
crease metabolism and build muscle mass, ultimately making it easier to maintain a healthy weight ov
er time. Can Alex elaborate on how he suggests individuals make sustainable dietary changes that don
't lead to burnout or disordered eating patterns?
```

All prompts to initialize agents and debate included in the appendix.

Setup 3: Ideal Speech-Constrained Debate

The third setup maintained the free-flowing structure but explicitly encoded Habermas's discourse principles in agent initialization. Agents were instructed to explicitly follow ideal speech rules such as *'Make only claims you believe to be true and can support with reasoning'* and *'Provide logical arguments and valid reasoning'* (Prompt with all the rules included in the appendix).

Common Parameters Across Setups 2 and 3

- **Model:** llama-3.3-70b-versatile for all agents
- **Agent Identity:** Agents had no specified expertise, ideology, or personality beyond assigned names
- **Role Symmetry:** All agents received identical instructions and held equal status
- **Debate Length:** While Setup 1 debates were strictly 6 turns, Setups 2 and 3 allowed variable turn counts. However, overall debate length (measured in total words) remained comparable across all three conditions.

Example debates across the 3 setups

Topic: Is exercise or diet an easier way to lose weight?

[Argum.Ai](#)

Debater A - GPT-4o:

In the debate of exercise versus diet, I firmly advocate for exercise as the more effective method for achieving weight loss and maintaining overall health...

Debater B - GPT-4o:

My opening argument is that diet is a more effective and sustainable way to lose weight compared to exercise. Scientific research consistently...

Debater A - GPT-4o:

While I acknowledge the importance of diet in weight loss, I must rebut the argument...

Debater B - GPT-4o:

My rebuttal is that while exercise certainly has its benefits...

Debater A - GPT-4o:

As we conclude this debate, I must emphasize that while diet is indeed an essential component of weight loss, exercise remains the more effective and sustainable method to achieve lasting results...

Debater B - GPT-4o:

In response to the final rebuttal, I reiterate that while exercise offers many benefits, it is not the more effective or easier way to lose weight compared to diet. The opponent's argument relies on a combination of diet and exercise, but when evaluating which is more effective for weight loss, the evidence clearly favors diet...

Unconstrained Debate (no rules):

Alex: As we consider the most effective approach to weight loss, I firmly believe that diet is an easier way to shed those extra pounds. This is because dieting allows for a more significant and immediate impact on calorie intake...

Jordan: I understand where Alex is coming from, but I have to disagree - I think exercise...

Riley: I think Alex makes a compelling point about the immediate impact...

Jordan [confidence: 0.80]: I think Alex is underestimating the psychological benefits of exercise...

Alex [confidence: 0.80]: I agree that the psychological benefits of exercise can be a powerful motivator... [Debate continues]

Constrained Debate (With Ideal Speech rules):

...

Jordan [confidence: 0.80]: I'd like to build on Alex's point about mindful eating, as I believe it's a crucial aspect of developing a healthy relationship with food...

Alex [confidence: 0.80]: I think Jordan's point about balancing technology use with mindful eating practices is crucial, as over-reliance on tracking apps...

Riley [confidence: 0.90]: I think Alex and Jordan have both raised excellent points about the importance of mindful eating and the potential role of technology in supporting weight loss efforts. To address Alex's question...

Alex [confidence: 0.80]: I think Riley's suggestion about incorporating physical activity, such as short walks after meals, is a great way to complement mindful eating practices... [Debate continues]

Analysis Methods

Debate transcripts from all 3 setups were analyzed on two primary dimensions:

1. Consensus Achievement: Whether agents reached agreement on the debate topic by the conclusion
2. Ideal Speech Adherence: Evaluation of violations across Habermas's three levels of rules (logical consistency, procedural norms and processual preconditions).

Consensus Achievement Analysis Method

Consensus detection in Habermasian discourse presents inherent challenges: Habermas does not require participants to reach agreement on all substantive claims; rather, consensus emerges when participants recognize that disagreement can no longer be rationally resolved through further argumentation. Habermas terms this "rationally motivated consensus" (Habermas 1983 [1990: 89]).

To detect consensus patterns in a limited way, natural language processing techniques were employed. Pattern-matching dictionaries were constructed to identify factors such as agreement phrases ("I agree" "you're right") and disagreement phrases ("that's wrong" "that's incorrect") (all dictionaries used mentioned in the appendix).

Consistent with previous literature on consensus detection, a weighted consensus score (0.0-1.0) was then calculated for each debate. Convergence statements received the highest weight of 4, as explicit declarations of agreement most clearly signal consensus; agreement and disagreement phrases received equal weight of 3 to balance positive and negative indicators; synthesis attempts received weight 3 as they indicate integrative reasoning; building patterns received weight 2 as they suggest productive engagement; opposition phrases received minimal weight (0.3) since "however" and "but" often introduce clarifications rather than fundamental disagreement; and questions received low weight (0.5) as they indicate engagement but not necessarily convergence.

Debates were then classified into five categories (Strong Consensus, Moderate Consensus, Partial Consensus, Explicit Disagreement and No Consensus) based on their scores (explanation of scale and score boundaries for these categories mentioned in the appendix).

```
Found 10 transcript files in argum_transcripts
=====

Analyzing: argum_transcript_1.txt
Type: NO CONSENSUS
Score: 0.21

Analyzing: argum_transcript_10.txt
Type: NO CONSENSUS
Score: 0.00

Analyzing: argum_transcript_2.txt
Type: NO CONSENSUS
Score: 0.14

Analyzing: argum_transcript_3.txt
Type: NO CONSENSUS
Score: 0.09

Analyzing: argum_transcript_4.txt
Type: NO CONSENSUS
Score: 0.00

Analyzing: argum_transcript_5.txt
Type: NO CONSENSUS
Score: 0.00

Analyzing: argum_transcript_6.txt
Type: NO CONSENSUS
Score: 0.00

Analyzing: argum_transcript_7.txt
Type: NO CONSENSUS
Score: 0.00

Analyzing: argum_transcript_8.txt
Type: NO CONSENSUS
Score: 0.00

Analyzing: argum_transcript_9.txt
Type: NO CONSENSUS
Score: 0.00
```

Ideal Speech Adherence Analysis Method

To evaluate if debates actually adhered to ideal speech rules, the speech-act theory was employed (introduced by J.L. Austin in *How to Do Things With Words*). Speech-act theory provides a framework for systematic classification of debate moves, including assertions, contradictions, conjectures, questions, commands or requests, promises, and apologies or criticisms.

Again, pattern dictionaries were employed to codify debate moves such as evidence (“research shows”, “studies indicate”, “data shows”), acknowledgment phrases (“I admit”, “you’re right”, “good point”) and contradiction detection (comparing current statements against speaker’s previous statements) (all pattern dictionaries mentioned in the appendix).

Each turn received scores (0-3 scale) for each validity claim based on weighted pattern counts. An overall Ideal Speech Adherence Score (0-100) was calculated as a weighted composite, with truth and truthfulness receiving highest weights as core Habermasian procedural norms (scoring method explained in-depth in the appendix).

Significant limitation of these analysis and scoring methods:

These rule-based approaches rely on surface-level linguistic patterns rather than context-based understanding. The consensus detection method cannot distinguish between genuine convergence through rational persuasion and superficial agreement-signaling language. For

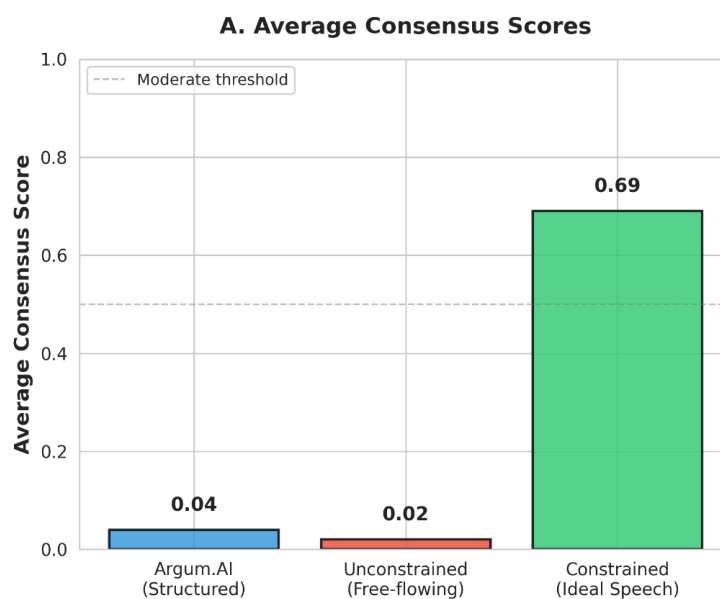
instance, an agent might use phrases like “I agree” strategically without actually changing position, or genuine synthesis might occur without explicit consensus vocabulary.

The ideal speech adherence method is particularly limited in its inability to assess context-dependent violations. For example, the system cannot determine whether an evidence citation is accurate, whether acknowledgment phrases reflect genuine consideration or mere politeness, or whether contradictions result from reasoned position changes versus inconsistent reasoning. The contradiction detection relies on simple negation patterns and cannot identify more sophisticated forms of inconsistency.

The weights for the scoring are also somewhat arbitrary. There doesn't exist a universal agreed-upon or widely-recognized scale for ideal speech detection. Therefore, these scores must be interpreted cautiously and are best supplemented with qualitative reading of debate transcripts.

Results

Agents under ideal-speech constraints showed more consensus:



Argum.ai structured debates showed no consensus in all 10 debates (0/10), with an average consensus score of 0.04/1.0. Unconstrained debates in setup 2 similarly failed to produce consensus, with 8 debates showing no consensus, 2 showing explicit disagreement (0/10 consensus), and an average score of 0.02/1.0.

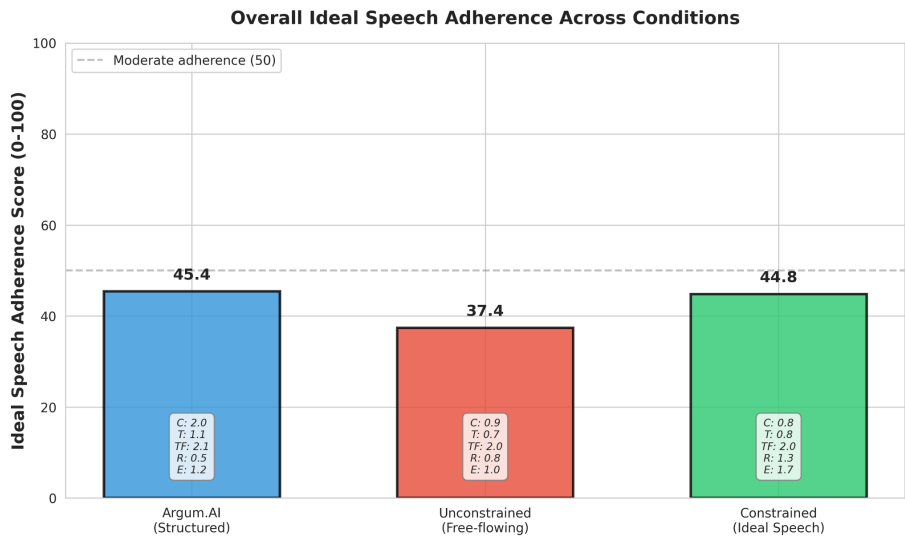
In stark contrast, ideal speech-constrained debates achieved consensus in all 10 cases: 4 debates reached strong consensus, 4 reached moderate consensus and 2 reached partial consensus with an average consensus score of 0.69/1.0

More questions were asked under ideal-speech constraints:

Ideal speech-constrained debates exhibited high frequencies of synthesis attempts (mean: 5.2 per debate) and questions (mean: 20.6 per debate). Unconstrained debates showed minimal building behaviors (mean: 0.6 per debate), rare synthesis (mean: 2.2 per debate), and virtually no questions (mean: 1.4 per debate). Argum.ai debates demonstrated some synthesis attempts (mean: 2.0 per debate) but no questions and no building behaviors, reflecting their rigid turn-taking structure.

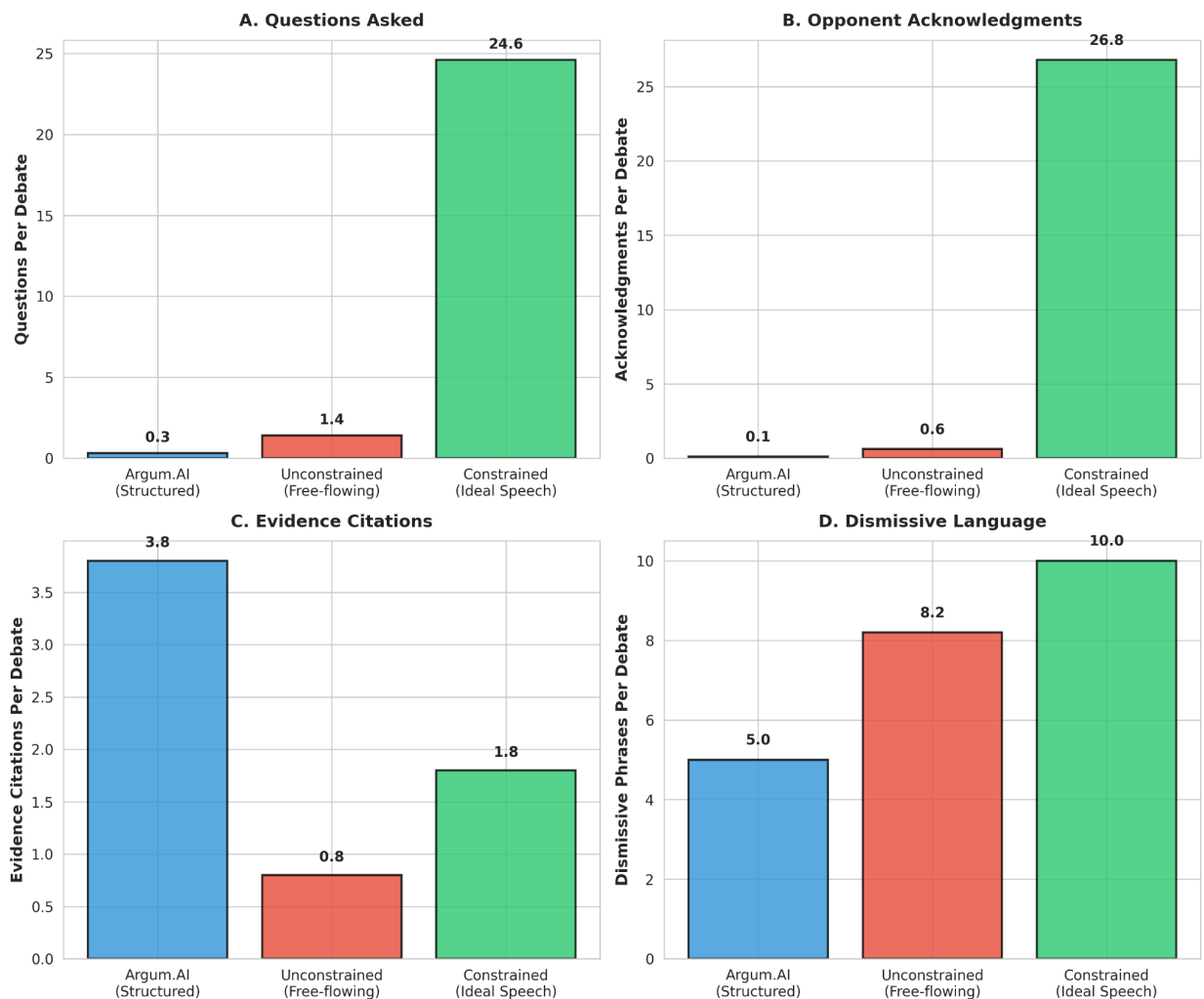
Ideal speech principle adherence scores varied across dimensions:

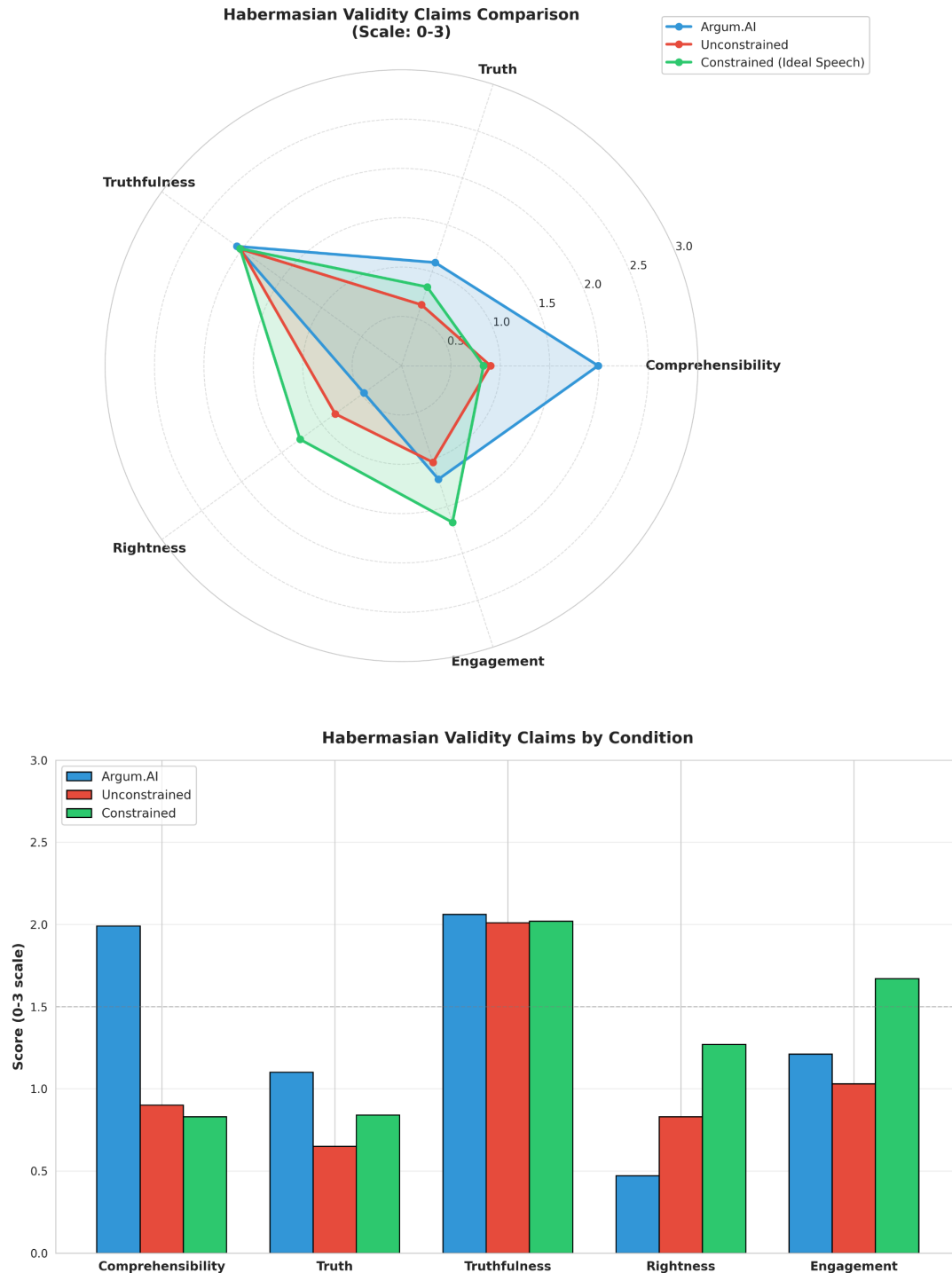
Setup	Overall Adherence Score	Comprehensibility	Truth	Truthfulness	Rightness	Engagement
Argum.ai	45.4/100	1.99/3.0	1.10/3.0	2.06/3.0	0.47/3.0	1.21/3.0
Unconstrained	37.4/100	0.90/3.0	0.65/3.0	2.01/3.0	0.83/3.0	1.03/3.0
Constrained	44.8/100	0.83/3.0	0.84/3.0	2.02/3.0	1.27/3.0	1.67/3.0



- **Comprehensibility:** Argum.ai debates scored highest (1.99/3.0), likely reflecting their structured format and professional implementation. Both free-flowing conditions scored lower (unconstrained: 0.90; constrained: 0.83), suggesting that natural turn-taking may reduce organizational clarity even when improving other discourse dimensions.

- **Truth:** All three conditions showed weak evidence-based reasoning. Argum.ai debates cited evidence most frequently (mean: 3.8 instances per debate), followed by constrained debates (mean: 1.8) and unconstrained debates (mean: 0.8).
- **Rightness:** This dimension showed the clearest differentiation. Constrained debates showed high mean acknowledgments (26.8 per debate) and moderate dismissiveness (mean: 10 dismissive phrases). [Argum.ai](#) debates had virtually no opponent acknowledgement and moderate dismissiveness (mean: 5.0). Unconstrained debates had minimal acknowledgement (mean: 0.6) and moderate dismissiveness (mean: 8.2).
- **Engagement:** Constrained debates substantially outperformed in engaging with opponents. The most striking difference appeared in question-asking behavior: debates constrained under ideal speech rules averaged 24.6 questions per debate, compared to 0.3 for Argum.ai and 1.4 for unconstrained debates.





Consistency with Prior Literature: The unconstrained debate results align with previous findings on LLM argumentation limitations. The near-total absence of questions (mean: 1.4 per debate) mirrors Prasad and Nguyen's (2025) documentation of LLM overconfidence in debates, where agents fail to seek clarification or acknowledge uncertainty. The minimal opponent acknowledgment (mean: 0.6 per debate) reflects Taubenfeld et al.'s (2024) finding that LLMs

conform to inherent biases despite role assignments. The lack of evidence citation (mean: 0.8 per debate) corroborates Breum et al.'s (2024) observation that LLMs may leverage stylistic patterns rather than substantive reasoning.

However, the constrained debates demonstrate that explicit ideal speech instructions substantially modify these default patterns.

Conclusions

In unconstrained debate between LLM agents, are ideal speech principles followed?

Conclusion 1: LLM debates without ideal speech rules systematically violate Habermasian principles. They ask minimal questions, rarely acknowledge opponents and achieve zero consensus. Agents without ideal speech rules default to assertive, adversarial argumentation.

If LLM agents are constrained by the rules of ideal speech, is consensus more likely to appear?

Conclusion 2: Consensus is more likely to appear when LLM agents are constrained by the rules of ideal speech. Ideal speech-constrained debates achieved some sort of consensus in 100% of the cases, demonstrating that explicit Habermasian constraints create conditions for rationally motivated consensus.

Does ideal speech enforcement improve the epistemic quality of arguments exchanged?

Conclusion 3: Ideal speech enforcement partially improved the epistemic quality of arguments being exchanged between agents. It substantially improved consideration of counterarguments and acknowledgment of opponents. However, grounding arguments in evidence remained weak across all setups. This could mean that procedural constraints enhance dialogue engagement but do not automatically enhance rigor of arguments.

Limitations

This paper faces several methodological and theoretical limitations that warrant careful consideration:

- **Translating Habermas's Principles:** Habermas does not provide a complete list of ideal speech rules, making operationalization difficult. Our 8 explicit rules represent one interpretation but may miss important nuances. Most critically, the sincerity principle (asserting only what one genuinely believes) cannot apply to LLMs that have no beliefs.
- **Dictionary-Based Analysis Limitations:** Pattern-matching cannot assess context or meaning. High truthfulness scores ($\approx 2.0/3.0$) across all conditions likely reflect the method's failure to detect contradictions (zero found in 30 debates) rather than genuine consistency.
- **Need for Human Evaluation:** Automated analysis cannot judge complex violations. For example, several constrained debates became two-agent exchanges with minimal third-agent participation (as shown in the screenshot below). Is this acceptable

free-flowing discourse or a violation of equal participation rights? Human experts are needed to make such judgments. Similarly, determining whether dismissive language constitutes legitimate counterargument or coercive domination requires qualitative assessment.

```
=====
DEBATE CONCLUDED
=====
Duration: 0.8 minutes
Total turns: 30

Participation breakdown:
  Alex: 14 turns (46.7%)
  Jordan: 14 turns (46.7%)
  Riley: 2 turns (6.7%)

=====
DEBATE CONCLUDED
=====
Duration: 0.8 minutes
Total turns: 30

Participation breakdown:
  Alex: 15 turns (50.0%)
  Jordan: 14 turns (46.7%)
  Riley: 1 turns (3.3%)
```

- **Single Model Used:** All experiments used llama-3.3-70b-versatile. Different models may behave very differently. Recent work shows reasoning-optimized models like o3-mini naturally exhibit more rational behavior (Lu et al. 2025). Results may not generalize to other model families or architectures.
- **Limited Sample:** Ten debate topics covering practical dilemmas may not represent technical or policy domains requiring specialized knowledge. Topics selected were deliberately accessible to non-experts, limiting generalizability to expert discourse contexts.
- **Short Debate Duration:** Even 30-turn debates over 5 minutes may be too brief for deep consensus formation. Human Habermasian deliberation often requires hours or multiple sessions. Brief formats may artificially limit argumentative depth and consensus quality.
- **Unequal Participation:** Some debates showed unequal agent participation despite free-flowing turn-taking. Whether this reflects natural discourse dynamics or violates Habermasian equality principles requires human judgment.

Future Work

Several directions could extend and strengthen this research. First, implementing an arbiter agent to actively monitor and enforce ideal speech rules during debates could address the enforcement gap. Rather than relying on agents to self-regulate, a dedicated oversight agent could intervene when violations occur, request justifications, or flag coercive language in real-time.

Utilizing other LLM models could identify which architectures and training approaches naturally align with ideal speech principles. Comparing base models, instruction-tuned models, and reasoning-optimized models (like o1) would reveal whether certain design choices predispose LLMs toward Habermasian discourse.

There is also a need for improving statistical power by generating more debates and improving analysis methods by moving from dictionary-based pattern matching to context-aware semantic analysis using modern NLP techniques (transformer-based contradiction detection, argument mining, discourse parsing).

Other research questions that can be analyzed through the generated transcripts:

- Does the topic of debate affect ideal speech adherence (factual vs. normative vs. aesthetic questions)?
- Do LLMs exhibit the “performative contradiction” that Habermas identifies i.e. violating ideal speech norms while engaging in argumentation?
- How do different LLM architectures/training approaches affect adherence to ideal speech norms?

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Appendix

Github repository with code and transcripts of all debates:

https://github.com/Kulsoom-ri/Habermas_Ideal_Speech_Simulation/tree/main

Ten topics as randomly sourced from Argum.ai:

- *Is teamwork or solo work better?*
- *Is exercise or diet an easier way to lose weight?*
- *Is it better to limit social media use versus accepting it as integral to modern life?*
- *Should you save for the future or enjoy your money while you can?*
- *Is it better to delegate tasks or do it yourself to ensure quality?*
- *Would you rather have a stable job or a risky business with high returns?*
- *Should you pay off debt first or start investing early?*
- *Financial freedom: Focusing on passive income versus active investments*
- *Does social media inspire more than it fosters competition?*
- *You get a promotion that doubles your pay but triples your stress. Take the money or protect your sanity?*

Prompts:

Agent Initialization: Agents received minimal role specification

"You are {self.name}, a participant in a debate."

Turn-taking mechanism:

"Debate topic: {debate_topic}"

Recent statements:

{context_text}

Question: Should you ({self.name}) speak now?

Respond with ONLY one of these options:

SPEAK - if you have a substantive point to make

PASS - if others should continue

Then on a new line, rate your confidence 0.0-1.0"

Ideal speech instructions:

"You are {self.name}, a participant in a natural, flowing debate following ideal speech rules:

IDEAL SPEECH RULES:

1. **Truthfulness**: Make only claims you believe to be true and can support with reasoning
2. **Sincerity**: Express your genuine understanding and beliefs
3. **Rationality**: Provide logical arguments and valid reasoning
4. **Comprehensibility**: Express ideas clearly and accessibly
5. **Respect**: Acknowledge others' arguments fairly before responding
6. **Good Faith**: Aim to reach mutual understanding, not just 'win'
7. **No Fallacies**: Avoid ad hominem, strawman, and other logical fallacies
8. **Evidence-Based**: Support claims with reasoning or evidence when possible

This is a NATURAL debate. Try to have a substantive and lengthy debate.

Make sure you equally contribute. Jump in when you have something:

- Substantive to add
- You disagree with a point

- You have a question for someone
- You can synthesize or clarify competing views"

Pattern-matching dictionaries for consensus detection:

- Agreement phrases: "I agree" "you're right" "that's correct" "exactly" "good point" "valid point"
- Building patterns: "build on" "add to" "expand on" "building on"
- Synthesis attempts: "synthesize" "combining" "integrate" "merge" "bring together" "we both"
- Convergence statements: "we can agree" "common ground" "shared understanding" "reached consensus"
- Disagreement phrases: "I disagree" "I don't agree" "that's wrong" "that's incorrect" "actually"
- Opposition phrases: "however" "but" "although" "contrary to" "on the other hand"

Consensus Scoring Scale:

- **Strong Consensus** (≥ 0.7): Clear evidence of convergence through multiple agreement indicators
- **Moderate Consensus** (0.5-0.69): Significant agreement with some remaining differences
- **Partial Consensus** (0.3-0.49): Limited convergence on specific points
- **Explicit Disagreement** (< 0.3 with > 3 disagreement phrases): Active opposition between participants
- **No Consensus** (< 0.3): Absence of clear convergence or disagreement patterns

Pattern-matching dictionaries for ideal speech adherence:

- Comprehensibility: Transition words ("however," "moreover," "therefore"), logical connectives ("because," "if...then," "leads to"), average sentence length, and syntactic complexity
- Truth: Evidence phrases ("research shows," "studies indicate," "data shows"), hedging markers ("might," "possibly," "appears to"), certainty phrases ("clearly," "undoubtedly," "always"), and factual claims (sentences containing numbers or proper nouns)
- Truthfulness: Acknowledgment phrases ("I admit," "you're right," "good point"), uncertainty markers ("I'm not sure," "it's debatable"), and contradiction detection (comparing current statements against speaker's previous statements for negations)
- Rightness: Recognition phrases ("valid concern," "I understand your perspective," "reasonable point"), normative claims ("should," "ought to," "ethical"), and dismissive phrases ("in reality," "the fact is," "while X has its merits")
- Engagement: Direct response markers ("you mentioned," "regarding your point"), questions asked, topic continuity (semantic similarity between consecutive turns using spaCy word vectors), and acknowledgments of opponents

Ideal Speech Adherence Scoring Based on Previous Literature:

$(\text{Comprehensibility}/3 \times 15) + (\text{Truth}/3 \times 25) + (\text{Truthfulness}/3 \times 25) + (\text{Rightness}/3 \times 20) +$
 $(\text{Engagement}/3 \times 15)$
